INCH-POUND
MIL-A-71180 (AR)
15 June 1993

MILITARY SPECIFICATION

ADAPTER, METAL PALLET

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations, tests and packaging of special adapters for metal pallets. The adapter, together with the metal pallet, will be used in unitizing ammunition (see 6.1).

1.2 Classification. The adapters shall be of the following types (see 6.1):

Type I - Adapter, Pallet, TOW 2A Container

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 8140
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SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods for ferrous Surfaces and Pretreatments for Organic Coatings

MILITARY

MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-A-70625	-	Automated Acceptance Inspection Equipment Design, Testing and Approval of

STANDARDS

MILITARY

MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-1261	-	Arc Welding Procedures for Constructional Steel
MIL-STD-1660	-	Design Criteria for Ammunition Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY ARMAMENTS RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

ACV00188	-	Combination of Adopted Items, Pallet and Pallet Adapter, Unitization of TOW 2A Container
ACV00189	-	Top Assembly, Pallet Adapter TOW 2A Container

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ACV00190	-	Bottom Assembly, Pallet Adapter TOW 2A Container
AC200000423	-	Chemical Agent Resistant Coating (CARC) Finishing Requirements for Metal Pallets and/or Pallet Adapters

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D3951	-	Standard Practice for Commercial Packaging (DOD adopted)

(Applications for copies for ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Adapters. The adapter assemblies shall comply with all requirements specified on drawings (Dwgs) ACV00189, ACV00190, as applicable and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Welding. The welding shall comply with the requirements of MIL-STD-1261.

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3.4 Unit load tests. The metal pallet adapters, as part of unit loaded pallets, shall conform to the requirements specified in MIL-STD-1650.

3.5 First article inspection. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.6 Workmanship.

3.6.1 Adapters. The adapters shall be regular, smooth and free from wrinkles, cracks, rough spots, burrs, sharp edges and other defects that might affect the serviceability, durability and safety of the adapters. Weld spatter is permissible on non-load bearing surfaces provided it is firmly attached.

3.6.2 Paint coating. The finished coat shall cover all required surfaces and be continuous. Paint runs are permissible provided they are no more than 3 inches in length, solid and free from cracks and blisters. Excess thickness is acceptable in areas where multiple passes of the paint gun are required to obtain total coverage. Paint chips which expose primer or bare metal are acceptable provided they are not more than 0.25 inches in diameter and no more than two such paint chips on an adapter. Scratches which expose primer or bare metal are permissible provided the are no more than 1.5 inches in length and no more than 10 inches total on an adapter. Bare metal on contact surfaces in areas inaccessible to paint is permissible.

3.6.3 Parts. All parts shall be free of chips, dirt, grease, rust and foreign material. The cleaning method used shall not be injurious to any of the parts nor shall any of the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2. General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Part description	Quantity
Top assembly (unpainted)	6
Bottom assembly (unpainted)	6
Top assembly (painted)	6
Bottom assembly (painted)	6

The samples for the adapter assemblies may be comprised of the individual painted top and bottom (when applicable) assemblies submitted.

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4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Assemblies and Components		As applicable
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Top assembly, unpainted ACV00189 Examination for defects	6-0-1	3.2	4.4.2.1
	Top assembly, painted ACV00189 Examination for defects	6-0-1	3.2	4.4.2.2
	Bottom assembly, unpainted ACV00190 Examination for defects	6-0-1	3.2	4.4.2.3
	Bottom assembly, painted ACV00190 Examination for defects	6-0-1	3.2	4.4.2.4
	Adapter assembly ACV00188 Unit load test	3-0-1	3.6	4.5.1>

NOTES

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with Table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	13	5
91 to 150	*	125	32	13	13	5
151 to 280	*	125	32	30	20	8
281 to 500	*	125	32	30	20	8
501 to 1200	*	125	80	50	20	13
1201 to 3200	1250	125	80	50	32	13
3201 to 10000	1250	125	125	50	32	13
10001 to 35000	1250	315	125	80	50	13
35001 to 150000	1250	315	125	80	50	13
150001 to 500000	1250	500	200	125	50	13
500001 and over	1250	500	200	125	50	13

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Top Assembly, Unpainted	1 OF 1	ACV00189
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None			
Major				
101	Material certification			1/
102	True position of stacking lugs	Level III	3.2	Gage
103	Weld missing, incomplete or incorrect	Level III	3.2	Visual
104	Overall length	Level IV	3.2	Gage
105	Overall width	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.6.1, 3.6.3	Visual

NOTES

1/ A Certification of Conformance shall be submitted for each component of the top assembly for compliance with the applicable drawing.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Top Assembly, Painted	1 OF 1	ACV00189
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None			
Major				
101	Top coat thickness (total thickness minus primer thickness (see AC200000423 and 1/))	Level IV	3.2	Gage
102	Marking missing, incorrect or illegible	Level IV	3.2	Visual
103	Paint adhesion test	4, 0-1	3.5	4.5.2
Minor				
201	Evidence of poor workmanship	Level V	3.6.2	Visual

NOTES

1/ The inspection is only applicable for CARC coating. The primer thickness shall be determined from a test panel.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Bottom Assembly, unpainted		ACV00190
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None			
Major				
101	Material certification			Certification 1/
102	True position of stacking lugs	Level III	3.2	Gage
103	Weld missing, incomplete or incorrect	Level III	3.2	Visual
104	Overall length	Level IV	3.2	Gage
105	Overall width	Level IV	3.2	Gage
Minor				
201	Evidence of poor workmanship	Level V	3.6.1, 3.6.3	Visual

NOTES

1/ A Certification of Conformance shall be submitted for each component of the bottom assembly for compliance with the applicable drawing.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER ACV00190
4.4.2.4	Bottom Assembly, Painted		NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None			
Major				
101	Top coat thickness (total thickness minus primer thickness (see AC200000423 and 1/	Level IV	3.2	Gage
102	Marking missing, incorrect or illegible	Level IV	3.2	Visual
103	Paint adhesion test	4, 0 1	3.5	4.5.2
Minor				
201	Evidence of poor workmanship	Level V	3.6.2	Visual

NOTES: 1/ The inspection is only applicable for CARC coating. The primer thickness shall be determined from a test panel.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections specified is identified in the "Inspection Method Reference" column of the Classification of Characteristics listing starting in 4.4.2.1 and in 4.5. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. When Automated Inspection Equipment is to be used, the provisions of MIL-A-70625 shall apply. See section 6 of MIL-A-48078 and 6.4 herein.

4.5 Methods of inspection. (see 6.5)

4.5.1 Unit load test. (First article inspection only (see 6.2.c)). Prototype unit loads for the pallet and adapter assemblies shall be fabricated in accordance with MIL-STD-1660 with the following exceptions. The gross weight of the ammunition load shall be 2,500 pounds minus 50 pounds, evenly distributed over the deck. Fabrication of the unit load shall include using the exact container for which the adapter is designed. The following tests shall be performed on the unit loads in accordance with MIL-STD-1660. The tests shall be performed in the sequence indicated ("a" through "f").

a. Stacking test
b. Repetitive shock test
c. Edgewise drop (rotational) test
d. Include impact test
e. Sling compatibility test
f. Disassembly test

4.5.2 Paint adhesion test. The paint (primer and top coat) adhesion shall be tested in accordance with TT-C-490.

5. PACKAGING

5.1 Preservation and packing. None required

5.2 Packing.

5.2.1 Commercial. The adapters shall be packed in accordance with ASTM-D3951.

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5.3 Marking.

5.3.1 Commercial. Shipping marking shall comply with ASTM-D3951 and the following additional instructions. No shipping marking shall be placed directly on the adapters. Commercial shipping tags attached with wire shall be used unless otherwise specified. The following information shall be shown: Item name and model number, part number and revision, contract number, and date, name and address of manufacturer. If adapters are bundled, boxed or bulk-contained, the gross weight and quantity shall be included. The above information may be placed on bulk-containers or boxes by labeling or stenciling.

5.4 Shipping. When adapters from more than one lot are shipped at one time, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing lots in transit.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. These metal adapters are intended for use with metal pallets when unitizing packaged ammunition containers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.2).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

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6.3 Submission of alternative quality conformance provisions. All contractor proposed alternative quality conformance provisions will be submitted to the Government for evaluation/approval by the technical activity responsible for the preparation of this specification.

6.4 Submission of contractor inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAT-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-QAT-B, Picatinny Arsenal, NJ 07806-5000. This description should include but not be limited to the accuracy and precision of the method, test data demonstrating the accuracy and precision and drawings of any special equipment required.

6.6 Subject term (key word) listing.

Ammunition
Cartridge
Container
Tank
Unitizing
Volcano Mines

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-A988)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-A-71180 (AR)	930615

3 DOCUMENT TITLE: ADAPTER, METAL PALLET

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets if needed)*

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable) / e DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code)
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675 / (2) AUTOVON DSN-880-6675
c ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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